

Deploying an ELK stack with Terraform

This project defines an Elasticsearch, Logstash and Kibana 9.5.4 lab on VMware Cloud Director. All three applications run on one Ubuntu 22.04 VM. Terraform manages the VM and its network attachment; cloud-init prepares the guest account; setup-elk.sh installs the applications over SSH.

Logstash receives events and sends them to Elasticsearch. Kibana queries Elasticsearch and displays the data in a browser. Nginx sits in front of Kibana and provides the HTTP and HTTPS entry points.

Setting	Configuration
Operating system	Ubuntu 22.04 LTS
Virtual machine	8 vCPUs, 16 GiB RAM, 60 GiB disk
Elasticsearch heap	4 GiB
Guest account	devops, with an authorized SSH key and sudo access

1. Prepare the configuration

Install Terraform 1.5 or later and OpenSSH on the workstation. The VCD organization, VDC, catalog, network and storage profile must already exist. Place the Ubuntu cloud OVA at files/ova/jammy-server-cloudimg-amd64.ova. The workstation needs SSH access to the VM, and the VM needs outbound access to the Ubuntu and Elastic package servers.

```
Copy-Item terraform.tfvars.example terraform.tfvars
notepad terraform.tfvars
```

Enter the VCD endpoint and credentials, the assigned VM address, domain, SSH public key, private-key path and SHA-512 password hash in terraform.tfvars. Replace the example address and domain with the lab values. Keep this file local; it is excluded by .gitignore.

```
vm_cpus      = 8
vm_cpu_cores = 1
vm_memory    = 16384
vm_disk_size = 61440
```

These settings define the VM size. The memory and disk inputs use MiB. The private key must match the public key passed to cloud-init.

2. Create the infrastructure

Run the following commands in PowerShell from the project directory:

```
terraform init
terraform fmt -check
terraform validate
terraform plan -out=tfplan
```

init installs the providers selected in .terraform.lock.hcl. fmt checks formatting, and validate checks the configuration against provider schemas. plan reads the current state and saves the proposed changes in tfplan. Review the resource names, address, storage profile and any deletions before applying.

```
terraform apply tfplan
```

File	What it defines
versions.tf, providers.tf	Provider versions and the VCD connection
catalog.tf	Ubuntu OVA upload to the existing catalog
main.tf	vApp, organization-network attachment and VM
elk.tf	Optional certificate files and the SSH provisioner
outputs.tf	VM address, SSH command and application URLs

A new deployment uses five resources by default: the catalog template, vApp, network attachment, VM and provisioner. With `use_custom_tls = true`, two local certificate files bring the total to seven. Both `domain_cert` and `domain_key` must contain base64-encoded values in that mode.

Guest setup

main.tf passes the cloud-init template as the VM's user-data property. The template sets the hostname, creates the devops account, installs basic utilities and configures SSH. It accepts the supplied password hash as well as the public key. The account has passwordless sudo.

The SSH provisioner waits for the VM, copies files/elk to /tmp/elk and runs setup-elk.sh with sudo. The installer configures Elasticsearch, Kibana, Logstash and Nginx. The provisioner removes its temporary certificate files after the installer returns.

Keep Terraform state and saved plans private. They can contain passwords and private-key material even when Terraform hides a value in console output. The provider lock file belongs in Git so another workstation can use the same versions.

3. Elasticsearch, Kibana and Nginx

Elasticsearch

The installer configures a single-node cluster and allocates a 4 GiB JVM heap. Elasticsearch listens on loopback and the VM address. Authentication is enabled, and the API uses HTTPS on port 9200.

```
cluster.name: bhos-elk-cluster
node.name: kibana-elk
network.host: ["127.0.0.1", "${VM_IP}"]
http.port: 9200
discovery.type: single-node
xpack.security.enabled: true
```

VM_IP is resolved by the installer on the guest. The HTTP and transport TLS settings refer to the certificate keystores under /etc/elasticsearch/certs. The local CA is also copied to Kibana and Logstash so they can verify Elasticsearch.

Kibana

Kibana connects to Elasticsearch as kibana_system. Its password is read from the VM's credential file and written to the local configuration, which is readable by root and the Kibana service group.

```
server.port: 5601
server.host: "0.0.0.0"
server.name: "kibana.local"
server.publicBaseUrl: "https://kibana.local"
elasticsearch.hosts: ["https://127.0.0.1:9200"]
elasticsearch.username: "kibana_system"
elasticsearch.password: "${KIBANA_PASSWORD}"
elasticsearch.ssl.certificateAuthorities: [ "/etc/kibana/certs/http_ca.crt" ]
```

Nginx

Nginx forwards requests from ports 80 and 443 to Kibana on 127.0.0.1:5601. It passes the original host and client-address headers to Kibana. The HTTPS server uses the certificate and key in /etc/nginx/ssl.

Use one hostname consistently in DNS, Nginx server_name, Kibana server.publicBaseUrl and the certificate. The supplied installer uses kibana.local. When no certificate is supplied, it creates a self-signed certificate for the lab. A browser certificate must cover the selected hostname and be trusted by the client.

4. Logstash and credential handling

Logstash accepts Beats connections on port 5044 and newline-delimited JSON over TCP on port 5000. The filter places the daily index name in event metadata. The Elasticsearch output uses that value when it writes each event.

```
input {
  beats {
    port => 5044
  }
  tcp {
    port => 5000
    codec => json_lines
  }
}

filter {
  mutate {
    add_field => { "[@metadata][target_index]" => "logstash-%{+YYYY.MM.dd}" }
  }
}

output {
  elasticsearch {
    hosts => ["https://127.0.0.1:9200"]
    ssl_certificate_authorities => ["/etc/logstash/http_ca.crt"]
    ssl_enabled => true
    user => "elastic"
    password => "${ELASTIC_PASSWORD}"
    index => "%{[@metadata][target_index]}"
  }
}
```

ELASTIC_PASSWORD is a Logstash keystore entry, not a password stored in the pipeline file. The installer creates the keystore, adds the credential and gives the Logstash service ownership of its configuration.

Credentials and local files

On the first run, the installer creates random passwords for elastic and kibana_system. It keeps them in /etc/elk/credentials.env with permissions 0600 and reuses them on later runs. Package-install output goes to a restricted log because package scripts may print credentials.

Do not commit terraform.tfvars, state, saved plans, private keys, certificates or access logs. The public example file contains replacement values only. Any password that was exposed must be changed wherever it was used.

This configuration is intended for a lab network. It enables HTTP and direct Kibana access, permits SSH password authentication and uses the elastic account for Logstash. All services share one VM.

5. Check access and remove the lab

Connect from PowerShell using the address returned by Terraform:

```
$vmIp = terraform output -raw vm_ip
ssh "devops@$vmIp"
```

In the Ubuntu shell, check the guest resources and service state:

```
lsb_release -ds
nproc
free -h
lsblk -dn -o NAME,SIZE,TYPE
systemctl is-active elasticsearch kibana logstash nginx
systemctl is-enabled elasticsearch kibana logstash nginx
```

Compare the reported resources with the Terraform inputs. Each systemctl command returns one status per service, in argument order. Inspect journalctl -u SERVICE on the VM if a service does not start.

```
sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt \
  --fail -u elastic https://localhost:9200/
sudo ss -lnt
```

curl prompts for the password. Check the version in the API response; /_cluster/health reports shard health separately. A single-node cluster can be yellow when replicas have no second node available.

Port	Purpose
22	SSH
80, 443	Nginx HTTP and HTTPS
5601, 9200	Kibana and the Elasticsearch HTTPS API
5044, 5000	Logstash Beats and TCP JSON

Configure DNS or C:\Windows\System32\drivers\etc\hosts for the Kibana hostname. From PowerShell, use Test-NetConnection -ComputerName \$vmIp -Port 443 to check client reachability, then open Kibana in the browser. A listening socket on the VM does not confirm that routing and firewalls permit the connection.

Cleanup

When the lab is no longer needed, review the destruction plan before deleting resources:

```
terraform plan -destroy
terraform destroy
```

References

[Terraform: sensitive data](#)

[Elastic: Logstash keystore](#)

[Elastic: Elasticsearch 9.5.4](#)